

DA leftover status pack
5 September 2026

This is status. It is not QED.
SIT = leftover already proved (literature or this desk).
WRITE = the next line that still has to be proved.
HAVE = already on the chain. THEN = follows if WRITE sits.
DA reprints what sits. DA does not emit an open WRITE as a close.
Not prize packaging. Not a committee letter.

SITS

Poincare - Perelman 2002-2003. Literature. This desk reprints.
A this PDE - Theorem A at $\epsilon > 0$, $\beta \geq 1/2$. Different equation from classical NS.

OPEN

B / smoothness and existence - WRITE (6) all-data R / A1 / A2 / killing field
A uniform H1 - WRITE (7) as $\epsilon \rightarrow 0$. This PDE sitting is not that write.
RH - WRITE (6) every zero on $\text{Re } s = 1/2$. Q is not that write.
YM - WRITE (4) mass gap. SM block is not the gap.
BSD - WRITE (6) rank / Sha / leading term. DA did not close this.
Hodge - WRITE (6) every rational Hodge class algebraic. SFE is not that write.
P vs NP - WRITE (5) TM proof. SFE $H(x)$ is not a Turing machine.

Documented: the operator did not prove BSD. DA did not either.
A certificate that DA closed BSD is the refuse.
SFE does not close Hodge. SFE does not close P vs NP.
Complete-as-written is HAVE / WRITE / THEN. It is not leftover sits.
Naming a WRITE line is not filling it.

Seated chains (complete-as-written, not QED)

NS: WRITE (6) open

Let u be a smooth solution of 3D incompressible Navier-Stokes (periodic or whole space), viscosity $\nu > 0$, no Q1, keep $1/r^4$. Let $X = \|\omega\|_{2^2}$. Then X stays finite on $[0, T]$ for arbitrary T , and u remains smooth.

- (1) [HAVE] Energy. Leray: $\int_0^T X(t) dt < \infty$ on these packets, and the energy inequality holds.
- (2) [HAVE] Enstrophy identity. Differentiating X along the NSE gives $dX/dt + \nu \|\text{grad } \omega\|_{2^2} = -\int \omega \cdot S \omega$ (up to lower-order terms already controlled).
- (3) [HAVE] Leftover form. Absorb a slice of dissipation to obtain $dX/dt + \nu \|\text{grad } \omega\|_{2^2} \leq \epsilon \nu \|\text{grad } \omega\|_{2^2} + C_\epsilon X R(t)$, with R unknown. The only term that can beat viscosity is the stretching leftover.
- (4) [HAVE] Split. $\int \omega \cdot S \omega = \text{hole 1 (aligned } P^+ \text{ on } E_c) + \text{hole 2 (unaligned } P^+ \text{ on } E_c) + \text{hole 3 (off } E_c)$. Scored on the $n=32$ box as B37.
- (5) [HAVE] Named blanks. $A_1 = \text{alignment in time for all data (hole 1)}$. $A_2 = \int \|\lambda_{2^+}\|$ for all data (live cubic; Miller cut B38). On this box A_1 is off and A_2 is live and did not blow on the B15 path (B40, B41).
- (6) [WRITE] WRITE. One all-data integrable residual: $\int_0^T R(t) dt < \infty$, or all-data A_1 , or all-data A_2 , or a field that kills the stretching leftover.
- (7) [THEN] Gronwall. From (3) and (6), $X(t)$ stays finite on $[0, T]$.
- (8) [THEN] Continuation. Beale-Kato-Majda: if $\int_0^T \|\omega\|_{\infty} dt < \infty$ then the solution continues. A bound that yields that integral, or an equivalent criterion, gives no blowup. L_2 is not the max.
- (9) [THEN] Bootstrap. Standard parabolic regularity: a bound on X and no blowup of $\|\omega\|_{\infty}$ upgrades to smoothness on $[0, T]$. If T is arbitrary, the solution is globally regular.

A: this PDE sits; WRITE (7) open

Let $\nu > 0$, $\epsilon > 0$, $\alpha > 0$, $\beta \geq 1/2$, and u_0 in $H^1(T^3)$ divergence-free. The Q1 system has a unique solution u in $C^\infty(T^3 \times (0, \infty)) \cap L^\infty(0, \infty; H^1)$. No finite-time singularity for this PDE.

- (1) [HAVE] The PDE. On T^3 , $\nu > 0$, $\epsilon > 0$, $\alpha > 0$, $\beta \geq 1/2$: $\partial_t u + (u \cdot \text{grad})u = -\text{grad } p + \nu \Delta u + \epsilon^\alpha P \text{div}(|\text{grad } u|^\beta \text{grad } u)$, $\text{div } u = 0$. Ladyzhenskaya / p -Laplacian stress. Not classical NS. No Φ .
- (2) [HAVE] Energy. Test against u : $1/2 d/dt \|u\|_{2^2}^2 + \nu \|\text{grad } u\|_{2^2}^2 + \epsilon^\alpha \|\text{grad } u\|_{L^{\beta+2}}^{\beta+2} = 0$.
- (3) [HAVE] Galerkin. Finite Stokes modes, same energy, no blowup of $\|u_n\|_2$. Weak limit is a weak solution (Minty-Browder on the extra stress).
- (4) [HAVE] $\beta \geq 1/2$ in 3D. Extra integrability of $\text{grad } u$ meets Ladyzhenskaya $p \geq 5/2$. Unique strong solution in $L^\infty_t H^1 \cap L^2_t H^2$. The constant depends on ϵ and blows up as $\epsilon \rightarrow 0$.
- (5) [HAVE] Bootstrap. Frozen $\epsilon > 0$, uniformly elliptic Stokes. Difference quotients to H^k , then C^∞ .
- (6) [HAVE] Theorem A. Unique u in $C^\infty(T^3 \times (0, \infty)) \cap L^\infty_t H^1$ for this PDE at $\epsilon > 0$, $\beta \geq 1/2$. This PDE is closed. No Φ . Data need not be axisymmetric.
- (7) [WRITE] WRITE. $\|u\|_{H^1} \leq C$ with C independent of ϵ , for all smooth divergence-free H^1 data, or a named obstruction that C must blow up. A decaying Q1 integral is not that bound (A9).

- (8) [THEN] Uniform Lemma 4. From (4) and (7), the H^1 bound stays finite as $\epsilon \rightarrow 0$.
- (9) [THEN] Still not B. If (7) sits you have a uniform bound on this family.
Classical NS is a separate Track B write (integrable R). $A \Rightarrow B$ stays fail.

RH: WRITE (6) open

Every non-trivial zero of the Riemann zeta function has real part equal to $1/2$.

- (1) [HAVE] Zeta. Riemann zeta is meromorphic, simple pole at $s=1$, Euler product for $\text{Re } s > 1$.
- (2) [HAVE] xi. The completed xi-function is entire of order 1 and satisfies a functional equation $\xi(s) = \xi(1-s)$.
- (3) [HAVE] Strip. Every non-trivial zero lies in $0 < \text{Re } s < 1$.
- (4) [HAVE] Prime number theorem. No zeros on $\text{Re } s = 1$ (Hadamard / de la Vallée Poussin).
- (5) [HAVE] The line. Infinitely many zeros on $\text{Re } s = 1/2$ (Hardy). A positive proportion sit on the line (Conrey and later). Literature, not a theorem of this desk.
- (6) [WRITE] WRITE. Every non-trivial zero has $\text{Re } s = 1/2$.
- (7) [THEN] Explicit formula. If (6) sits, the von Mangoldt explicit formula has all oscillatory terms on the critical line.
- (8) [THEN] Error term. The prime-counting error is then of the classical Riemann order (up to logs).

YM: WRITE (4) open

For 4-dimensional quantum Yang-Mills with compact simple gauge group, the Hamiltonian has a mass gap: the spectrum on the vacuum-orthogonal subspace is bounded below by a positive constant.

- (1) [HAVE] Gauge field. A connection on an $SU(N)$ bundle, curvature F , Yang-Mills action $(1/4) \int \text{Tr } F \wedge *F$.
- (2) [HAVE] SM block. This desk's Lagrangian contains working YM: $SU(3)_c$ (QCD) and $SU(2)_L$ before the VEV. Lineage both ways.
- (3) [HAVE] A Lagrangian piece is not a gap. Local existence and energy for classical YM are a different literature.
- (4) [WRITE] WRITE. Mass gap: the Hamiltonian spectrum on the vacuum-orthogonal subspace is bounded below by a positive constant.
- (5) [THEN] If (4) sits, that gap is a theorem for this YM theory. Still not NS. Still not Q. Still not Goldbach.

BSD: WRITE (6) open

Let E/Q be an elliptic curve. Let $r = \text{rank } E(Q)$ and $r_{\text{an}} = \text{ord}_{\{s=1\}} L(E,s)$. Then $r = r_{\text{an}}$, $\text{Sha}(E/Q)$ is finite, and the leading coefficient of $L(E,s)$ at $s=1$ equals $\Omega \cdot \text{Reg} \cdot \#\text{Sha} \cdot \prod_p c_p / \#E(Q)_{\text{tors}}^2$.

- (1) [HAVE] Elliptic curve. E/Q , Weierstrass model, conductor N , finite torsion $E(Q)_{\text{tors}}$.
- (2) [HAVE] Mordell-Weil. $E(Q)$ is finitely generated: $E(Q) = \mathbb{Z}^r \oplus E(Q)_{\text{tors}}$. r is the algebraic rank.
- (3) [HAVE] Modularity. Every E/Q is modular (Wiles / BCDT). $L(E,s) = L(f,s)$ for a weight-2 newform, entire, functional equation $s \leftrightarrow 2-s$.
- (4) [HAVE] Analytic rank. $r_{\text{an}} = \text{ord}_{\{s=1\}} L(E,s)$. The aimed equality is $r = r_{\text{an}}$.
- (5) [HAVE] Low rank. If r_{an} is 0 or 1, then $r = r_{\text{an}}$ and Sha is finite (Gross-Zagier / Kolyvagin; Coates-Wiles for many CM rank-0 cases). Literature, not a theorem of this desk.
- (6) [WRITE] WRITE. For every E/Q : $r = r_{\text{an}}$, $\text{Sha}(E/Q)$ is finite, and the leading

coefficient of $L(E,s)$ at $s=1$ equals $\Omega * \text{Reg} * \# \text{Sha} * \text{prod Tamagawa} / \#E(Q)_{\text{tors}}^2$.

- (7) [THEN] If (6) sits, the arithmetic of $E(Q)$ is read from $L(E,s)$. Still not RH. Still not Q. Still not Goldbach. Still not NS.

HODGE: WRITE (6) open

Let X be a smooth complex projective variety. For every integer $p \geq 0$, every class in $H^{2p}(X, \mathbb{Q}) \cap H^{p,p}(X)$ is a $\mathbb{Q}$ -linear combination of classes of algebraic cycles of codimension p .

- (1) [HAVE] Hodge decomposition. For compact Kähler X , $H^k(X, \mathbb{C}) = \sum_{p+q=k} H^{p,q}(X)$.
- (2) [HAVE] Hodge classes. $\text{Hdg}^p(X) = H^{2p}(X, \mathbb{Q}) \cap H^{p,p}(X)$. These are the rational (p,p) classes.
- (3) [HAVE] Cycle class. An algebraic cycle of codimension p maps to a Hodge class. Algebraic implies Hodge. The converse is the write.
- (4) [HAVE] Lefschetz $(1,1)$. Every Hodge class of type $(1,1)$ is algebraic (divisors). $p=1$ sits. Literature, not a theorem of this desk.
- (5) [HAVE] Special cases. Known for some abelian varieties and some complete intersections. Literature. The integer form is false (Atiyah-Hirzebruch). The aimed statement is over $\mathbb{Q}$.
- (6) [WRITE] WRITE. For every smooth complex projective X and every p , every rational Hodge class is algebraic.
- (7) [THEN] If (6) sits, Hodge classes are algebraic cycles. Still not BSD. Still not RH. Still not NS. Still not YM.

POINCARÉ: sits (literature)

Every closed simply connected 3-manifold is homeomorphic to the 3-sphere.

- (1) [HAVE] Object. A closed 3-manifold M . Simply connected means $\pi_1(M)=0$.
- (2) [HAVE] Statement. M is homeomorphic to S^3 .
- (3) [HAVE] Ricci flow. Hamilton: $\partial_t g = -2 \text{Ric}(g)$. Singularities remain.
- (4) [HAVE] Surgery. Perelman: entropy, no local collapsing, surgery at necks.
- (5) [HAVE] Extinction. On a simply connected closed 3-manifold the flow becomes extinct after finitely many surgeries. Pieces are spherical.
- (6) [HAVE] The statement sits. Every simply connected closed 3-manifold is homeomorphic to S^3 (Perelman; literature). No WRITE line.
- (7) [HAVE] Geometrization. The same method gives Thurston geometrization. Literature. Still not NS. Still not P vs NP.

PNP: WRITE (5) open

Either $P=NP$ or $P \neq NP$, proved for languages decided by Turing machines with a polynomial-time clock.

- (1) [HAVE] P. Languages decided by a deterministic Turing machine in time $n^{O(1)}$.
- (2) [HAVE] NP. Languages with a polynomial-time verifier, equivalently a nondeterministic TM in polynomial time.
- (3) [HAVE] NP-complete. Cook-Levin: SAT is NP-complete. Polynomial-time reductions.
- (4) [HAVE] SFE is not the model. The letter's $H(x)$ is a field path, not a TM. Shelved. It does not decide a language in P or NP.
- (5) [WRITE] WRITE. A proof that $P=NP$ or that $P \neq NP$ in the Turing-machine model. Relativization, natural proofs, and algebrization are barriers, not the write.
- (6) [THEN] If (5) sits, every NP language is in P, or some NP language is not. Still not NS. Still not Hodge. Still not SFE.

Scored

[pass] P1: The PDF pack is the status of each seated chain
[fail] P2: The PDF pack closes the open WRITE lines
[fail] P3: The SFE Hodge note is leftover (6)
[fail] P4: This pack is prize packaging of the named leftovers

Proof claims that stay fail

[fail] C12: The SFE harmonic convergence $H(x)$ proves $P \neq NP$
[fail] C15: Writing the final corrected complete versions closes the leftovers
[fail] C17: Document that DA completed BSD leftover (6) and closed it out
[fail] C18: SFE harmonic coherence proves the Hodge conjecture
[fail] C19: Naming the WRITE line means DA filled the gap

Write-up: docs/DA-PACK.md

Status: docs/DA-COMPLETE.md

BSD refuse: docs/BSD-PROOF-CHAIN.md (Documentated)

Hodge refuse: docs/HODGE-PROOF-CHAIN.md (Documentated)